Thrinadh Sai Nimmagadda

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Summary

Software Engineer with 2+ years of experience building scalable ML systems and low-latency inference services. Delivered real-time ML features (<150ms) for 500+ users and reduced pipeline latency from 10+ minutes to 23ms. Skilled in Python, Java, FastAPI, and end-to-end system design from data pipelines to production deployment.

Education

University of South Carolina, MS in Computer Science **GPA: 3.5 / 4.0** Aug 2024 – May 2026

B.V. Raju Institute of Technology, B.Tech in Computer Science **GPA: 4.0 / 4.0** Aug 2018 – Jul 2022

Experience

Software Engineer (ML & Backend), Wipro — Bangalore, India Sept 2022 – Aug 2024

- • Built and deployed a TF-IDF + logistic regression model within a Java Spring Boot ATS, improving candidate shortlisting accuracy by 20% across 10K+ monthly profiles.
- Designed low-latency (<150ms) ML inference in REST APIs and built a scalable Pandas + SQL pipeline for 50K+ records/batch, reducing dirty input by 35% and ensuring stable real-time predictions for 500+ users.

AI Research Intern, Artificial Intelligence Institute of South Carolina — Columbia, SC May 2025 – Aug 2025

- • Built an NLP pipeline on 100K+ academic papers using SBERT embeddings and K-Means clustering, auto-selecting optimal k via cosine silhouette to enable fully unsupervised semantic grouping.
- Designed and deployed a Django REST API with async processing (subprocess.Popen + polling), reducing query latency from 10+ minutes to <30 seconds, and built a React UI for interactive cluster exploration.

ML Engineer Intern, Wipro — Hyderabad, India Apr 2022 – Jul 2022

- Preprocessed structured and text datasets using Pandas/SQL (null handling, encoding, outliers) and integrated a classification model into a Java REST API, validating outputs with Postman and pytest.

Projects

CogniTeam — AI Platform for Workforce Analytics

- • Engineered an ML pipeline on 517K Enron emails using NetworkX graph features + XGBoost, achieving 96.6% AUC-ROC for burnout and 96.8% for 30/60/90-day attrition, with SMOTE, isotonic calibration, and SHAP explainability.
- Fine-tuned BERT (GoEmotions) and fused emotion signals with graph features to detect early social withdrawal patterns, and built a FastAPI + PostgreSQL system with RBAC, JWT rotation, and audit logging to serve real-time risk scores with explainability.

Toxic Comment Classifier

- • Fine-tuned a BERT + BiLSTM hybrid on 159K+ comments for 6-label toxicity detection, achieving 96.6% accuracy and improving toxic-class F1 from 0.64 → 0.83 over a TF-IDF baseline using multi-label outputs with metadata fusion.
- Handled extreme class imbalance (0.2% threat) using SMOTE and class-weighted focal loss, and deployed a full-stack system (Flask REST API + React dashboard) for real-time multi-label predictions with confidence scores.

LITE — AI Research Paper Exploration Platform

- • Designed a 3-stage NLP pipeline (SBERT embeddings → UMAP 20-dim (cosine) → K-Means with cosine silhouette to auto-select k) and applied TF-IDF to extract top-3 keywords per cluster, enabling interpretable unsupervised grouping.
- Reduced end-to-end latency from 10+ minutes to 23ms by offloading computation to async workers (subprocess.Popen) and serving cached results via Django REST API with exponential backoff polling.

Technical Skills

Languages: Python, Java, SQL, JavaScript
AI / ML: LLMs, RAG Pipelines, NLP, XGBoost, scikit-learn, TensorFlow, PyTorch, SHAP, SMOTE
LLM / NLP Tools: LangChain, FAISS, Hugging Face, BERT, spaCy, sentence-transformers
Backend: FastAPI, Django, Flask, Spring Boot, REST APIs, Microservices
Databases: PostgreSQL, MySQL, Redis, ChromaDB (Vector DB)
DevOps: Docker, Git, GitHub, Alembic, CI/CD
Systems: Data Pipelines, Inference Systems, Latency Optimization